

Bosun

- CLOUD-NATIVE PLATFORM FOR HOSTING AND SCHEDULING CODING AGENT -

Group
Six / Cloud
Computing



Our Live Demo

Motivation

- Initial Motivation:** AI coding agents are interactive, stateful and resource-intensive. Running them on personal machines ties sessions to one device, provides limited multi-user isolation and makes computing resources difficult to manage efficiently.
- So Our Overview:** Bosun moves these workloads to a shared Kubernetes cluster. It preserves each user with workspace while providing controlled isolation, centralized lifecycle management and adaptive resource allocation.

Advantages

Multi-Tenant Isolation

Each user receives a dedicated Kubernetes Namespace protected by resource, network and permission policies. ResourceQuota and LimitRange control resource allocation, NetworkPolicy restricts communication, and RBAC limits Kubernetes operations.

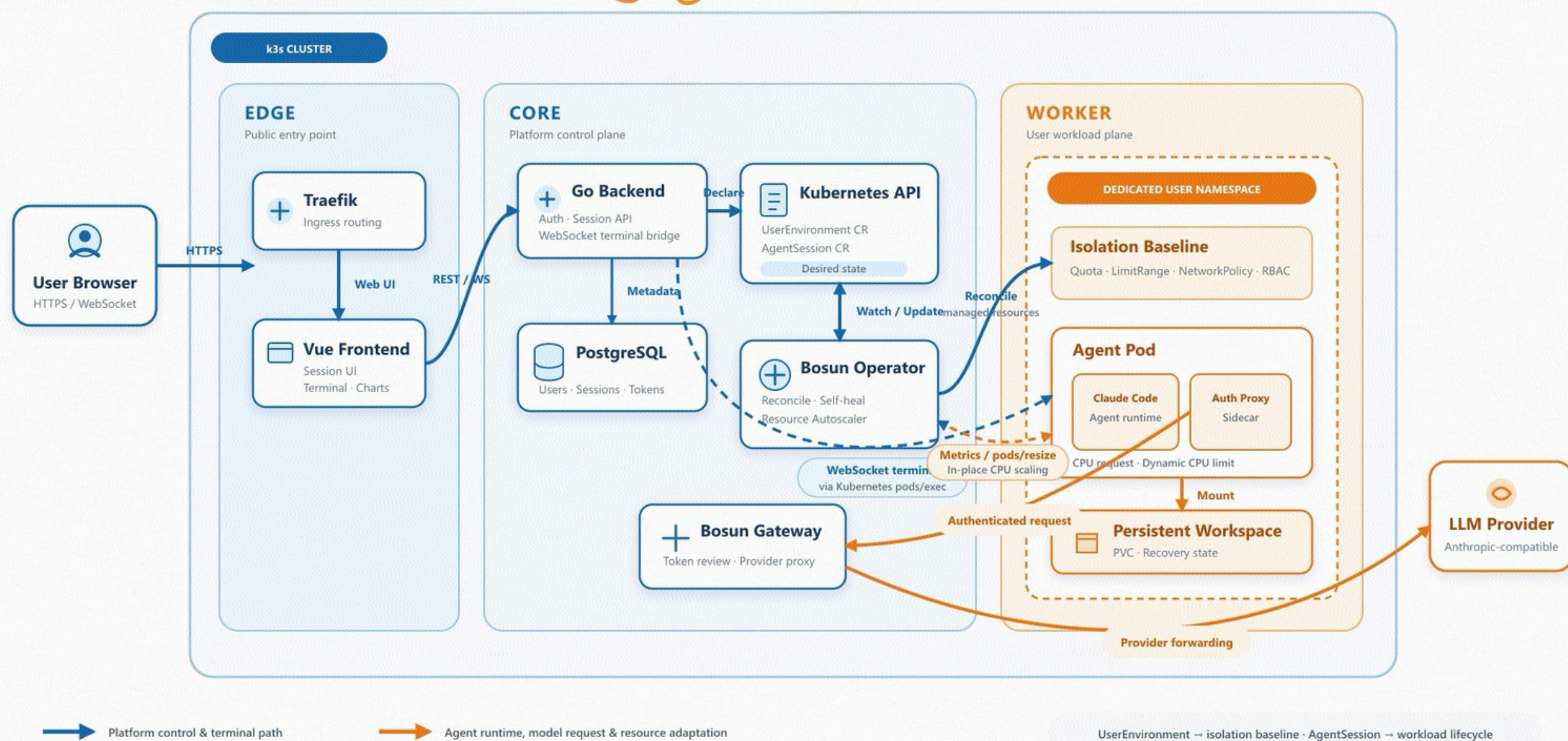
Persistent Agent Lifecycle

Each session has a PVC-backed workspace and an Operator-managed lifecycle. Sessions can be created, connected, hibernated, resumed and deleted. Hibernation removes the Agent Pod to release compute while retaining the user's workspace and recovery state.

Adaptive Resource Management

Node places Agent Pods on worker nodes, while non-preempting PriorityClass levels order waiting sessions when capacity is limited. Live metrics drive in-place CPU vertical scaling, allowing the Agent container's CPU limit to change without replacing the running Pod.

Architecture

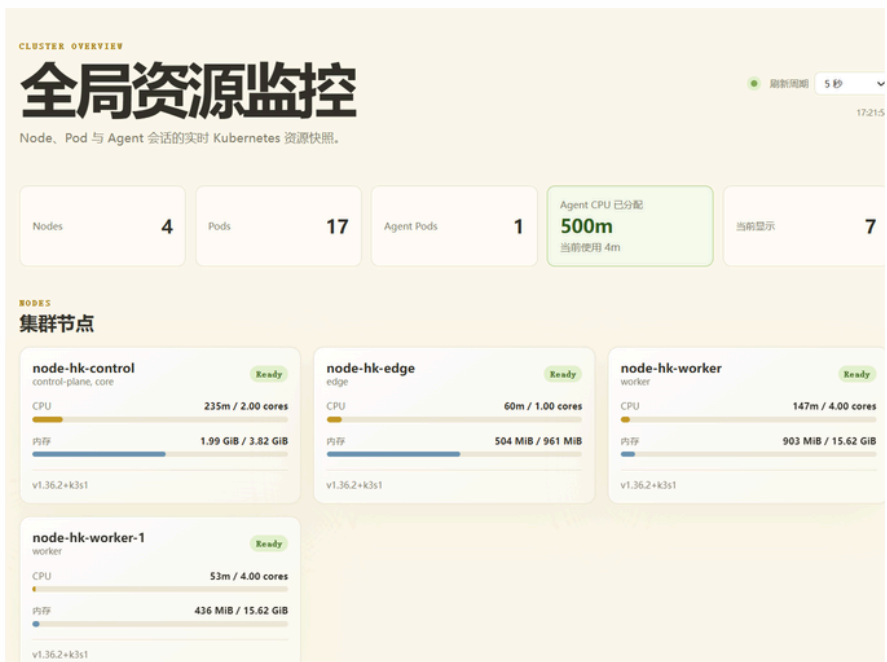


Tools



Platform in Action

- Schedule, Observe & Scale:** Track Pod placement and live resource usage across the cluster. Priority-aware scheduling orders waiting sessions, while in-place vertical scaling adjusts CPU limits without restarting active Pods.



- Create & Manage:** Create Agent sessions, select resource and priority settings, and manage lifecycle states through the web interface.

- Code in the Browser:** Access a persistent Claude Code terminal directly from the browser without exposing the Agent Pod publicly.



github.com/Amsors/Bosun

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